

EA Credit: Advanced Energy Metering

This prerequisite applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core & Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses & Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To support energy management and identify opportunities for additional energy savings by tracking building-level and system-level energy use.

REQUIREMENTS

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Install *advanced energy metering* for the following:

- ▶ all whole-building energy sources used by the building; and
- ▶ any individual energy end uses that represent 10% or more of the total annual consumption of the building.

The advanced energy metering must have the following characteristics.

- ▶ Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.
- ▶ Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- ▶ The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- ▶ The system must be capable of storing all meter data for at least 36 months.
- ▶ The data must be remotely accessible.
- ▶ All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

CS

Install meters for future tenant spaces so that tenants will be capable of independently metering energy consumption (electricity, chilled water, etc.) for all systems dedicated to their space. Provide a sufficient number of meters to capture total tenant energy use with a minimum of one meter per energy source per floor.

Install *advanced energy metering* for all base building energy sources used by the building.

The advanced energy metering must have the following characteristics.

- ▶ Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.
- ▶ Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- ▶ The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- ▶ The system must be capable of storing all meter data for at least 36 months.
- ▶ The data must be remotely accessible.

- ▶ All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Submetering energy subsystems helps facility managers track changes in energy usage over time and provides the performance data necessary to identify opportunities for energy savings by end use. Submetering is an important component of a successful energy management program; metered data enables monitoring of consumption and costs as well as progress reporting throughout the building life cycle.